

LIGHT & SHADOW

Two Souls. One Heist.

Phase 2 – Requirement Analysis & Feasibility Review

*Project-Based Placement Readiness Programme
Training and Placement Cell*

Team Number: XV | Team / Repository Name: Light And Shadow
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01. Project Proposal (Recap)

This section restates the Phase 1 project proposal for context, in line with the Team01_LightAndShadow/01_Project_Proposal.pdf requirement.

Abstract

LIGHT & SHADOW is a premium top-down, puzzle-stealth heist game for PC and consoles. Building on the responsive movement and tactile level design of Robbery Bob, the game introduces an original dual-character traversal mechanic built entirely around the physical manipulation of light and shadow. Players control two spectral thief entities: Light, who can only walk within illuminated beams, and Shadow, who can only traverse dark, unlit areas. A light beam is a physical path for one character and an impassable wall for the other. By manipulating dynamic light sources, rotating mirrors, and refraction physics (prisms and lenses), players coordinate both characters to bypass guard systems, steal high-value loot, and escape each level.

Problem Statement

- Binary Stealth Predictability — in most stealth games, light only affects visibility, not traversal, producing static "wait for the guard to turn" gameplay.
- Passive Environmental Interaction — puzzle-stealth environments are usually static; players rarely construct their own navigation paths.
- Asymmetric Co-op Limitations — cooperative puzzle games rarely give two characters fundamentally different navigation rules, where one player's floor is the other's hazard.

Proposed Solution

A stylised, top-down puzzle-stealth experience in which players switch between, or cooperatively control, two characters governed by mutually exclusive navigation rules. Light can move only inside illuminated beams; Shadow can move only within unlit areas. Traversal meshes recalculate in real time as light sources swing, rotate, reflect, and refract, so players must actively sculpt the paths available to each character.

Key Features

- Dual-perspective cooperative play: single-player hot-swap with tactical pause, or local/online co-op.
- Real-time dynamic NavMesh generator driven by light raycasts for both characters.
- Physics-based optical puzzles: mirrors, lenses, filters, splitter prisms, and shadow-casting props.
- Stark premium aesthetic: high-contrast noir visuals, glassmorphic UI, adaptive soundtrack.
- Robbery Bob-inspired movement: grid-aligned controls, sneak/sprint speeds, tactile hiding spots.

02. Work Per Person – Phase 1 Contribution Summary

Individual contributions during the idea-submission phase (Phase 1), consistent with the Team01_LightAndShadow/02_Work_Per_Person.pdf requirement. Roles reflect each member's primary area of ownership; all members collaborated on ideation and review.

Team Member	Primary Contribution (Phase 1)	Carried Forward Into Phase 2
Alvin Jomon	Core concept design: dual-character light/shadow traversal rules, problem statement, and abstract.	Functional requirements & use-case definition.
Akshay	Mechanics design: mirror reflection, prism refraction, and shadow-casting	Technical feasibility & literature survey on optical puzzle design.

	puzzle systems.	
Ajas	Guard AI concept: Lumen Guard, Nyx Guard, and Dynamic Searchlight Guard behaviour design.	Non-functional requirements & AI-related feasibility analysis.
Michael	High-level architecture, technology stack research, and documentation formatting.	Schedule/economic feasibility & work allocation planning.

Note: precise task hours and commit references should be filled in from the team's repository log before final submission.

03. Requirement Analysis & Literature Survey

3.1 Functional Requirements

Core functionalities the system is expected to perform:

- FR1 — Provide two independently controllable characters, Light and Shadow, each restricted to movement within their respective terrain type (illuminated vs. shadowed).
- FR2 — Support hot-swapping between characters in single-player mode, including a tactical pause on swap.
- FR3 — Support local and online cooperative play with one character per player.
- FR4 — Recalculate a Light NavMesh and a Shadow NavMesh in real time from raycasts cast by all active light sources.
- FR5 — Simulate rotating/swinging light sources (spotlights, pendulum lanterns) that move over time on defined timers or physics.
- FR6 — Allow players to rotate, slide, and align mirrors on tracks to redirect light beams.
- FR7 — Simulate refraction through triangular prisms, splitting white light into red/green/blue beams that trigger matching colour-coded gates.
- FR8 — Allow Shadow to push or pull dynamic objects (crates, sliding walls, pillars) to cast new shadows ("shadow bridges").
- FR9 — Give Light a flare ability that blinds guards or activates light-sensitive sensors.
- FR10 — Give Shadow a cloaking ability and object-manipulation ability to block or create shadow paths.
- FR11 — Drive Lumen Guards, Nyx Guards, and Dynamic Searchlight Guards through a state machine (Patrol, Search, Chase, Blinded), each querying the NavMesh relevant to their vision type.
- FR12 — Trigger an alert or player reset when a character leaves its permitted NavMesh (e.g., Shadow enters light).
- FR13 — Provide a level editor for placing light sources, mirrors, prisms, guards, and loot.
- FR14 — Track win conditions (loot secured and both characters reach the extraction point) and loss conditions (detection/alarm triggered).

3.2 Non-Functional Requirements

Quality Attribute	Requirement
Performance	Maintain a stable 60 FPS on target platforms (Switch, PS5, Xbox Series X/S, low-spec PC) even during real-time light-recalculation and NavMesh updates.
Security	Protect online co-op sessions from tampering (e.g., speed-hacks, NavMesh spoofing) through server-authoritative state validation.
Reliability	Recover gracefully from a dropped co-op connection without corrupting level or save state.
Scalability	Level format and NavMesh generator must scale from small tutorial rooms to large multi-guard heist levels without redesign.
Usability	Traversal rules must be communicable through visual language alone (beam colour, shading) with minimal on-screen text.
Maintainability	Light physics, NavMesh generation, guard AI, and gameplay rules are decoupled into independent modules to allow isolated updates (see Phase-1 architecture diagram).
Portability	Built on Unity URP for cross-platform compilation across console and PC targets.

3.3 Literature Survey

Review of existing work relevant to dual-character stealth, cooperative heist design, and optical puzzle mechanics.

Reference	Relevant Technique	Limitation	Research Gap Addressed
Robbery Bob (level & sneak mechanics)	Responsive grid-aligned movement, immediate spatial stealth feedback, tactile hiding spots.	Light/dark states only modulate visibility, not traversal — movement rules stay identical everywhere.	LIGHT & SHADOW turns lighting into physical terrain, so traversal itself changes with lighting, not just detection odds.
Monaco: What's Yours is Mine (asymmetric heist co-op)	Class-based co-op traversal; each player perceives/interacts with the level differently.	Class differences are largely perceptual (vision filters) rather than physically exclusive movement zones.	LIGHT & SHADOW makes the two roles' movement domains mutually exclusive and spatially opposed, deepening the asymmetric co-op design space.
Portal (optical/physics puzzle design)	Reorienting vectors and rays to redirect energy/traversal paths through the environment.	Single-character, single traversal rule-set; no two-agent lighting conflict.	LIGHT & SHADOW extends ray-based puzzle logic to a two-agent system where the same beam is a path for one agent and a wall for the other.
Unity URP 2D Lighting documentation	Real-time 2D light/shadow rendering pipeline with custom render features.	Provides rendering only; no built-in gameplay-traversal coupling to light data.	Confirms technical feasibility of sourcing real-time illumination data to drive a custom NavMesh generator.
FMOD Studio interactive audio guide	State-based adaptive scoring driven by gameplay parameters.	Generic tooling; requires project-specific parameter design.	Used to justify feasibility of a soundtrack that crossfades between Light- and Shadow-themed instrumentation based on active character.

Overall, existing stealth and puzzle titles treat lighting as either a pure detection modifier (Robbery Bob) or a single-agent optical puzzle element (Portal), while asymmetric co-op titles (Monaco) differentiate roles perceptually rather than through mutually exclusive terrain. The identified research gap is a two-agent system in which the same physical light beam is simultaneously a walkway and a wall, depending on which character occupies it — this is the core innovation LIGHT & SHADOW proposes to fill.

04. Feasibility Study

4.1 Technical Feasibility

Unity 2022.3 LTS with URP provides native 2D light/shadow rendering, raycasting, and Shader Graph support, all of which the Light Physics Engine and Dynamic NavMesh Generator depend on. Mirror/prism reflection-refraction math is standard 2D vector geometry, well within reach of C# scripting. The main technical risk is real-time NavMesh regeneration cost as light sources move; this is mitigated by baking navigation on a coarse grid and only recomputing regions touched by a moving light source each frame, rather than the full level.

4.2 Economic Feasibility

All core tools proposed (Unity, Git/Git LFS, FMOD Studio indie tier) are free or low-cost for a student/indie team, and the project is scoped as an academic vertical-slice prototype rather than a full commercial release. Primary cost is development time across the four-member team rather than licensing; no specialised hardware beyond standard development PCs and console dev-kits (for later porting) is required for the prototype stage.

4.3 Operational Feasibility

The team already has defined role ownership (mechanics, AI, architecture/tech stack, documentation) from Phase 1, and the proposed architecture is modular, so members can work on the Light Physics Engine, NavMesh Generator, Guard AI, and Gameplay Controller in parallel with limited integration friction. Git with Git LFS supports the collaborative, asset-heavy workflow needed for textures, audio, and sprites.

4.4 Schedule Feasibility

The Expected Outcome in the Phase 1 proposal scopes deliverables into a 3-level vertical slice, a basic level editor, and cross-platform performance validation — a staged scope that fits an academic project timeline. Early phases (mechanics + single-character traversal) can be validated before the more complex refraction/colour-gate and multi-guard-AI systems are layered in, reducing schedule risk from any single subsystem.

05. Work Allocation Among Team Members

Proposed allocation of major project components for the development phase, building on each member's Phase 1 ownership area.

Team Member	Module Owned	Key Responsibilities
Alvin Jomon	Gameplay Controller & Rules	Implement traversal rule checks (Light/Shadow zone validation), win/loss evaluation, and level-editor integration for loot/objective placement.
Akshay	Light Physics Engine	Build raycasting, mirror reflection, and prism refraction systems; define the colour-coded gate logic.
Ajas	Guard AI State Machine	Implement Lumen Guard, Nyx Guard, and Dynamic Searchlight Guard behaviours (Patrol, Search, Chase, Blinded) against the relevant NavMesh.
Michael	Dynamic NavMesh Generator & Tooling	Build the Light/Shadow NavMesh generation pipeline from the polygon map, plus the developer-facing level editor and documentation.

Every member additionally contributes to shared tasks: co-op input handling, FMOD audio integration, and cross-platform performance profiling. Allocation should be revisited at each milestone as module dependencies become clearer.